

Video Link: <https://youtu.be/mefW5NIOx74>

## PowerBI Project

### Data Preparation and Cleaning:

#### *Ticket Information Dataset*

| TicketID | FlightID | BookingStatus  |
|----------|----------|----------------|
| 1        | 5001     | 1178 Pending   |
| 2        | 5002     | 1078 Confirmed |
| 3        | 5003     | 1117 Cancelled |
| 4        | 5004     | 1120 Cancelled |
| 5        | 5005     | 1137 Cancelled |
| 6        | 5006     | 1162 Pending   |
| 7        | 5007     | 1076 Pending   |
| 8        | 5008     | 1035 Cancelled |
| 9        | 5009     | 1001 Cancelled |
| 10       | 5010     | 1040 Cancelled |
| 11       | 5011     | 1064 Pending   |
| 12       | 5012     | 1150 Cancelled |
| 13       | 5013     | 1060 Cancelled |
| 14       | 5014     | 1064 Confirmed |
| 15       | 5015     | 1093 Confirmed |
| 16       | 5016     | 1072 Pending   |
| 17       | 5017     | 1011 Cancelled |
| 18       | 5018     | 1105 Cancelled |
| 19       | 5019     | 1014 Confirmed |
| 20       | 5020     | 1060 Pending   |
| 21       | 5021     | 1030 Confirmed |
| 22       | 5022     | 1035 Confirmed |
| 23       | 5023     | 1165 Confirmed |
| 24       | 5024     | 1005 Confirmed |

#### *Passenger Information Dataset*

| PassengerID | FlightID | SeatNumber |
|-------------|----------|------------|
| 1           | 1        | 1161 38A   |
| 2           | 2        | 1157 24D   |
| 3           | 3        | 1141 30B   |
| 4           | 4        | 1046 17E   |
| 5           | 5        | 1035 29D   |
| 6           | 6        | 1134 10A   |
| 7           | 7        | 1082 10A   |
| 8           | 8        | 1115 20E   |
| 9           | 9        | 1197 34E   |
| 10          | 10       | 1047 2E    |
| 11          | 11       | 1153 43C   |
| 12          | 12       | 1194 48C   |
| 13          | 13       | 1010 47A   |
| 14          | 14       | 1056 23C   |
| 15          | 15       | 1030 16D   |
| 16          | 16       | 1109 40D   |
| 17          | 17       | 1005 25C   |
| 18          | 18       | 1119 32C   |
| 19          | 19       | 1033 27E   |
| 20          | 20       | 1118 32B   |
| 21          | 21       | 1065 19E   |
| 22          | 22       | 1146 5B    |
| 23          | 23       | 1177 28B   |
| 24          | 24       | 1011 22E   |

Flight Information Dataset

|    | FlightID | FlightNumber | Airline   | Destination | Status    |
|----|----------|--------------|-----------|-------------|-----------|
| 1  | 1001     | FL1102       | Airline D | Houston     | On Time   |
| 2  | 1002     | FL1435       | Airline B | Chicago     | On Time   |
| 3  | 1003     | FL1860       | Airline A | New York    | Cancelled |
| 4  | 1004     | FL1270       | Airline C | Chicago     | Delayed   |
| 5  | 1005     | FL1106       | Airline C | New York    | Delayed   |
| 6  | 1006     | FL1071       | Airline A | Phoenix     | On Time   |
| 7  | 1007     | FL1700       | Airline C | Los Angeles | Cancelled |
| 8  | 1008     | FL1020       | Airline C | Los Angeles | Delayed   |
| 9  | 1009     | FL1614       | Airline A | Los Angeles | Cancelled |
| 10 | 1010     | FL1121       | Airline D | Chicago     | Cancelled |
| 11 | 1011     | FL1466       | Airline A | Phoenix     | On Time   |
| 12 | 1012     | FL1214       | Airline D | New York    | Delayed   |
| 13 | 1013     | FL1330       | Airline C | Houston     | On Time   |
| 14 | 1014     | FL1458       | Airline C | New York    | Delayed   |
| 15 | 1015     | FL1087       | Airline C | Houston     | Delayed   |
| 16 | 1016     | FL1372       | Airline B | New York    | Delayed   |
| 17 | 1017     | FL1099       | Airline D | Phoenix     | Delayed   |
| 18 | 1018     | FL1871       | Airline B | Houston     | Delayed   |
| 19 | 1019     | FL1663       | Airline B | Chicago     | Cancelled |
| 20 | 1020     | FL1130       | Airline A | New York    | On Time   |
| 21 | 1021     | FL1661       | Airline B | New York    | Cancelled |
| 22 | 1022     | FL1308       | Airline A | Houston     | Delayed   |
| 23 | 1023     | FL1769       | Airline A | Chicago     | On Time   |
| 24 | 1024     | FL1343       | Airline B | Chicago     | Delayed   |

Data Modeling:

Manage relationships

New relationshipAutodetect

EditDeleteFilter

| From: table (column) ↑             | Relationship | To: table (column)                  | Status |
|------------------------------------|--------------|-------------------------------------|--------|
| Passenger_Information - passe...   | * ← 1        | Flight_Information - flight_info... | Active |
| Ticket_Information - ticket_inf... | * ← 1        | Flight_Information - flight_info... | Active |

Enhanced Data Insights:

Table.RenameColumns("Inserted Text After Delimiter",{"Text After Delimiter", "Flight Number"}, {"Custom", "Condition"})

|    | FlightNumber | Airline | Destination | Status      | Condition | Flight Number  |
|----|--------------|---------|-------------|-------------|-----------|----------------|
| 1  | 1001         | FL1102  | Airline D   | Houston     | On Time   | Best           |
| 2  | 1002         | FL1435  | Airline B   | Chicago     | On Time   | Best           |
| 3  | 1003         | FL1860  | Airline A   | New York    | Cancelled | To Be Improved |
| 4  | 1004         | FL1270  | Airline C   | Chicago     | Delayed   | To Be Improved |
| 5  | 1005         | FL1106  | Airline C   | New York    | Delayed   | To Be Improved |
| 6  | 1006         | FL1071  | Airline A   | Phoenix     | On Time   | Best           |
| 7  | 1007         | FL1700  | Airline C   | Los Angeles | Cancelled | To Be Improved |
| 8  | 1008         | FL1020  | Airline C   | Los Angeles | Delayed   | To Be Improved |
| 9  | 1009         | FL1614  | Airline A   | Los Angeles | Cancelled | To Be Improved |
| 10 | 1010         | FL1121  | Airline D   | Chicago     | Cancelled | To Be Improved |
| 11 | 1011         | FL1466  | Airline A   | Phoenix     | On Time   | Best           |
| 12 | 1012         | FL1214  | Airline D   | New York    | Delayed   | To Be Improved |
| 13 | 1013         | FL1330  | Airline C   | Houston     | On Time   | Best           |
| 14 | 1014         | FL1458  | Airline C   | New York    | Delayed   | To Be Improved |
| 15 | 1015         | FL1087  | Airline C   | Houston     | Delayed   | To Be Improved |
| 16 | 1016         | FL1372  | Airline B   | New York    | Delayed   | To Be Improved |
| 17 | 1017         | FL1099  | Airline D   | Phoenix     | Delayed   | To Be Improved |
| 18 | 1018         | FL1871  | Airline B   | Houston     | Delayed   | To Be Improved |
| 19 | 1019         | FL1663  | Airline B   | Chicago     | Cancelled | To Be Improved |
| 20 | 1020         | FL1130  | Airline A   | New York    | On Time   | Best           |
| 21 | 1021         | FL1661  | Airline B   | New York    | Cancelled | To Be Improved |
| 22 | 1022         | FL1308  | Airline A   | Houston     | Delayed   | To Be Improved |
| 23 | 1023         | FL1769  | Airline A   | Chicago     | On Time   | Best           |
| 24 | 1024         | FL1343  | Airline B   | Chicago     | Delayed   | To Be Improved |
| 25 | 1025         | FL1491  | Airline D   | Phoenix     | On Time   | Best           |
| 26 | 1026         | FL1413  | Airline D   | Chicago     | Cancelled | To Be Improved |
| 27 | 1027         | FL1805  | Airline D   | Chicago     | On Time   | Best           |

Query Settings

PROPERTIES

Name

Flight\_Information - flight\_information.xlsx

All Properties

APPLIED STEPS

Source

Promoted Headers

Changed Type

Removed Duplicates

Removed Blank Rows

Added Conditional Column

Inserted Text After Delimiter

Renamed Columns

Calculations Using DAX Formula:

1. Total Passengers for a specific flight

| Count of PassengerID | FlightID |
|----------------------|----------|
| 1                    | 1001     |
| 2                    | 1003     |
| 1                    | 1004     |
| 1                    | 1005     |
| 1                    | 1006     |
| 1                    | 1010     |
| 1                    | 1011     |
| 1                    | 1012     |
| 1                    | 1023     |
| 1                    | 1026     |
| 1                    | 1027     |
| 2                    | 1030     |
| 1                    | 1032     |
| 1                    | 1033     |
| 1                    | 1034     |
| 1                    | 1035     |
| 1                    | 1038     |
| 1                    | 1039     |
| 1                    | 1046     |
| 2                    | 1047     |
| 2                    | 1050     |
| 1                    | 1051     |
| 1                    | 1052     |
| 1                    | 1053     |
| 1                    | 1054     |
| 1                    | 1055     |
| 1                    | 1056     |
| 1                    | 1057     |
| 1                    | 1059     |
| 1                    | 1063     |
| 1                    | 1065     |
| 1                    | 1066     |
| 100                  |          |

2. Total flights booked

```
1 TotalTicketsBooked= CALCULATE(COUNT('Ticket_Information - ticket_information'[TicketID]))
```

50

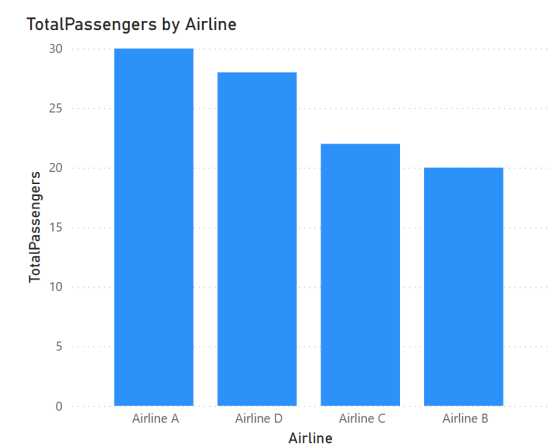
TotalTicketsBooked

3. Filtered Table showing “Best” flights only

| FlightNumber | Airline   | Destination | Status  | Condition | Flight Number |
|--------------|-----------|-------------|---------|-----------|---------------|
| 1001         | Airline D | Houston     | On Time | Best      | 1001          |
| 1002         | Airline B | Chicago     | On Time | Best      | 1002          |
| 1006         | Airline A | Phoenix     | On Time | Best      | 1006          |
| 1011         | Airline A | Phoenix     | On Time | Best      | 1011          |
| 1013         | Airline C | Houston     | On Time | Best      | 1013          |
| 1020         | Airline A | New York    | On Time | Best      | 1020          |
| 1023         | Airline A | Chicago     | On Time | Best      | 1023          |
| 1025         | Airline D | Phoenix     | On Time | Best      | 1025          |
| 1027         | Airline D | Chicago     | On Time | Best      | 1027          |
| 1028         | Airline D | Chicago     | On Time | Best      | 1028          |
| 1029         | Airline D | Los Angeles | On Time | Best      | 1029          |
| 1030         | Airline B | Phoenix     | On Time | Best      | 1030          |
| 1031         | Airline B | New York    | On Time | Best      | 1031          |
| 1033         | Airline D | New York    | On Time | Best      | 1033          |
| 1034         | Airline B | Phoenix     | On Time | Best      | 1034          |
| 1036         | Airline D | Phoenix     | On Time | Best      | 1036          |
| 1039         | Airline B | Chicago     | On Time | Best      | 1039          |
| 1043         | Airline C | Houston     | On Time | Best      | 1043          |
| 1044         | Airline B | Phoenix     | On Time | Best      | 1044          |
| 1046         | Airline D | Chicago     | On Time | Best      | 1046          |
| 1048         | Airline A | New York    | On Time | Best      | 1048          |
| 1050         | Airline C | Phoenix     | On Time | Best      | 1050          |
| 1052         | Airline D | Phoenix     | On Time | Best      | 1052          |
| 1053         | Airline C | Chicago     | On Time | Best      | 1053          |
| 1055         | Airline B | New York    | On Time | Best      | 1055          |
| 1057         | Airline A | Phoenix     | On Time | Best      | 1057          |
| 1060         | Airline D | Chicago     | On Time | Best      | 1060          |

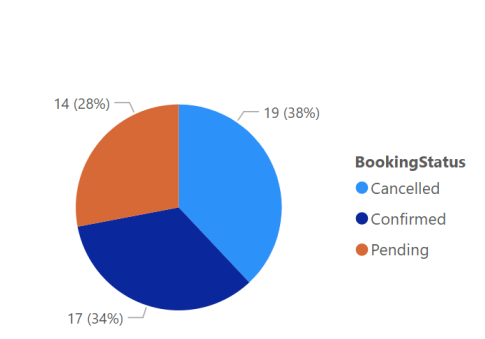
Visualization and Interactive Features:

1. Passenger count by airline



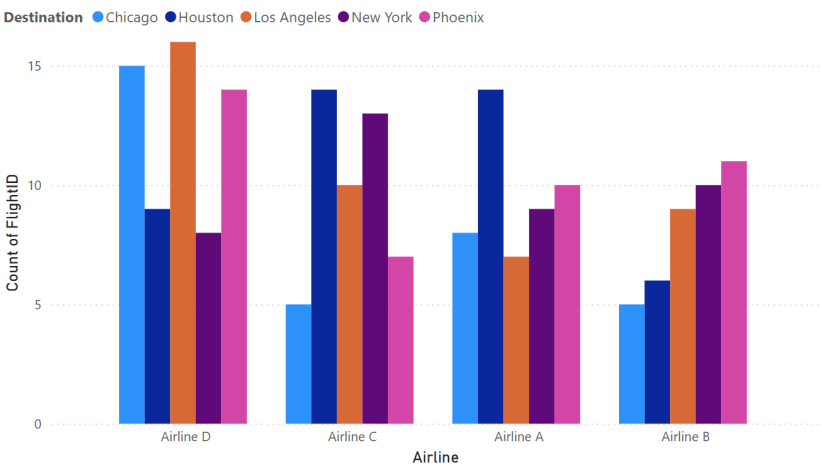
2. Ticket Booking Status

Count of TicketID by BookingStatus

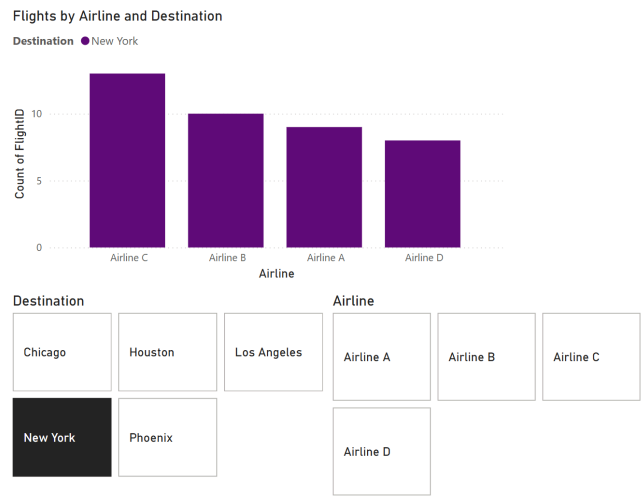


3. Flights by airline and destination

Flights by Airline and Destination



Interactive Features:



6

TotalPassengers

8

TotalFlights

2

TotalTicketsBooked

7

TotalPassengers

14

TotalFlights

3

TotalTicketsBooked

Airline, Destination

^ ▣ Airline A

- ▣ Chicago
- ▣ Houston
- ▣ Los Angeles
- ▣ New York
- ▣ Phoenix

✓ ▣ Airline B

✓ ▣ Airline C

✓ ▣ Airline D

Airline, Destination

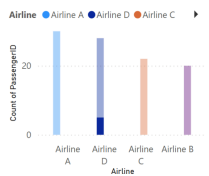
^ ▣ Airline A

- ▣ Chicago
- ▣ Houston
- ▣ Los Angeles
- ▣ New York
- ▣ Phoenix

^ ▣ Airline B

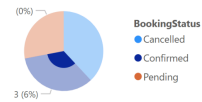
- ▣ Chicago
- ▣ Houston

Count of PassengerID by Airline and Airline

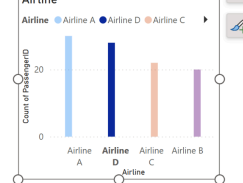


| Destination | Airline   |
|-------------|-----------|
| Chicago     | Airline A |
| Houston     | Airline A |
| Los Angeles | Airline A |
| New York    | Airline A |
| Phoenix     | Airline A |
| Chicago     | Airline B |
| Houston     | Airline B |
| Los Angeles | Airline B |
| New York    | Airline B |
| Phoenix     | Airline B |
| Chicago     | Airline C |
| Houston     | Airline C |
| Los Angeles | Airline C |
| New York    | Airline C |
| Phoenix     | Airline C |
| Chicago     | Airline D |
| Houston     | Airline D |
| Los Angeles | Airline D |
| New York    | Airline D |
| Phoenix     | Airline D |

Count of BookingStatus by BookingStatus



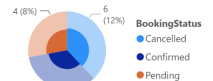
Count of PassengerID by Airline and Airline



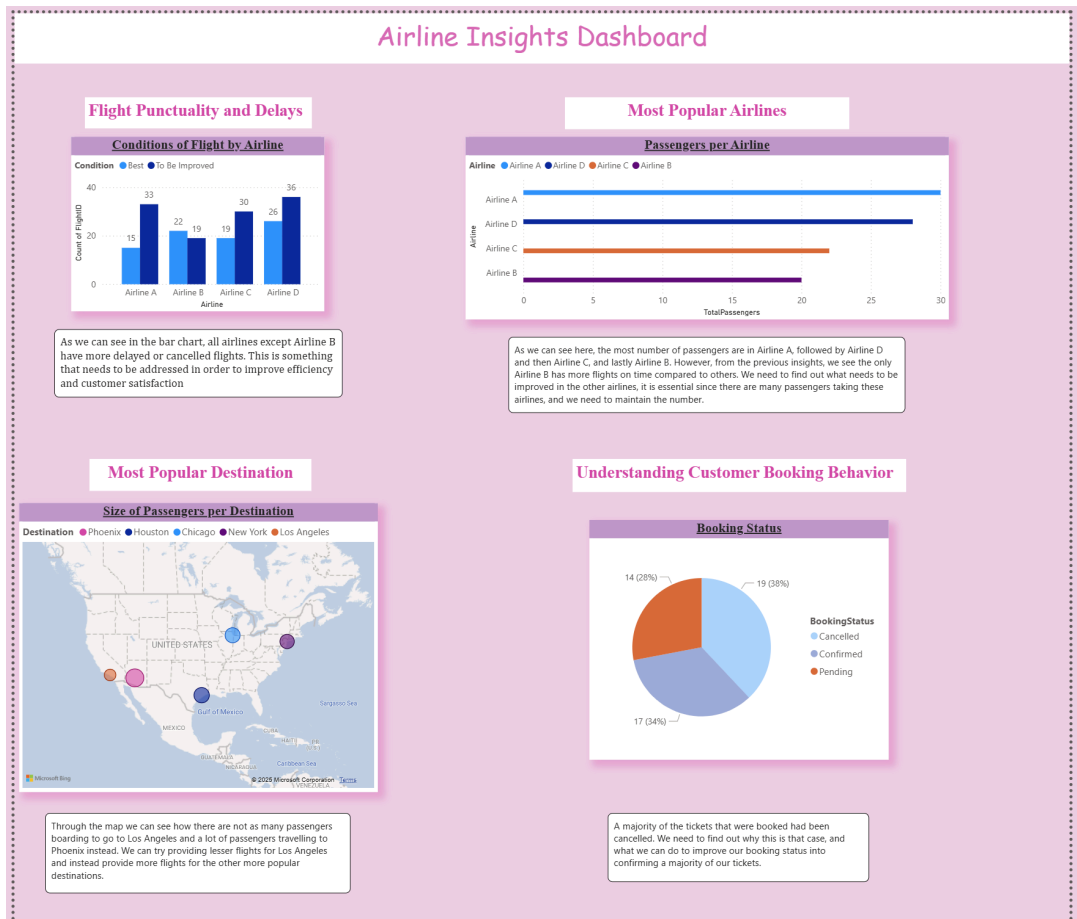
Destination Airline

|             |           |
|-------------|-----------|
| Chicago     | Airline D |
| Houston     | Airline D |
| Los Angeles | Airline D |
| New York    | Airline D |
| Phoenix     | Airline D |

Count of BookingStatus by BookingStatus



Final Dashboard and PowerBI Services:



Manage security roles



Create new security roles and use filters to define row-level data restrictions.

Successfully applied role changes.

+ New

Airline A Role

Flight\_Informa...

Passenger\_Info...

Ticket\_Informa...

Filter data

Switch to default editor

1 [Airline]= "Airline A"

Filter the data that this role can see by entering a DAX filter expression that returns a True/False value. For example: [Entity ID] = "Value"

